Error-tolerant Beam Steering of mmWave Antennas by Trajectory Estimation of Highway Vehicles

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Abstract— **We study the effect of information update delay on the minimum required beam width of the antenna for information exchange. For high-speed data information exchange, millimeter wave antennas are our choice of antenna and we deploy them along the highways to provide connectivity to the high-speed vehicles. We present a novel technique of predicting the trajectory of the vehicle for a short interval of time by approximation, which performs at par with the accurate calculations. The calculations also account for the Gaussian noise present in the GPS information which leads to an error in the predicted position. Considering the trade-off between the update delay and the directivity of the beam, we aim to obtain an optimum range of beamwidth by scaling and intersecting the directivity curve with update delay curve.**

I. Introduction

Infotainment services are widely consumed these days which require a high data rate. Millimetre wave links can provide a bit rate of up to 7 Gbps for vehicles moving at speed less than 100 kmph [1], which is more than enough to deliver high quality video content to the user. Along the highways connectivity is poor owing to the high mobility of the vehicles. Therefore, it is useful to have antennas which can track the vehicle by beam steering without much delay.

V2I communication is a wireless exchange of information between moving vehicles and stationary road objects. This technology is becoming more and more popular as it has significantly helped to cut down travel time as well as reduce traffic congestion. Since the critical information exchanged will be processed to ensure safe mobility, the data carrying capacity of the medium has to be large enough to support it along with a minimal delay.

As presented in [3] the deployed systems in Europe have ITS sub-systems which are capable of IEEE 802.11p wireless communication. This standard can only support a maximum bandwidth of 10 MHz with corresponding data rates of 3-27 Mbps. In our situation, such data rates are not sufficient. Hence we need to switch over to wideband communication involving millimeter waves.

While employing millimeter waves, there is a specific problem which needs to be addressed. These waves are directional unlike other low frequency waves which can transmit data in any direction provided the receiver is within bounds. Hence a proper steering of these beams is required to ensure that information is conveyed suitably to the vehicles.

Mavromatis [4] states that the problem of in-band overhead beam formation can be addressed by utilizing the information transmitted by the vehicles at any given instant. This data is used to predict the motion of the vehicles and hence the beamwidth can be appropriately altered to increase the throughput. This MAC-layer algorithm, SAMBA (Smart Motion-prediction Beam Alignment) also performs better than IEEE 802.11ad BF strategy and hence could be a feasible solution for the same.

II. Methodology

We consider a highway scenario where independent vehicles are sending their instantaneous GPS coordinates (x_0, y_0) along with their velocity vector (v_x, v_y) information to the millimetre wave antenna when a LOS connection is established. Based on this information, the short-term trajectory of the vehicle can be predicted and the antenna having maximum angular velocity can be aligned so as to provide connection to the vehicle.

We calculate the position of the vehicle , at time + Δ using the formula,

$$\tan(\theta_0 - \omega_{max}\Delta) = \frac{y_0 + v_y\Delta}{x_0 + v_x\Delta} \tag{1}$$

Small angle approximation gives us

$$\Delta = \frac{v_x \tan\theta_0 - v_y - x_0\omega_{max}\sec^2\theta_0}{\omega_{max}(v_x + v_y\tan\theta_0)} \tag{2}$$

where θ_0 is the LOS angle of current position of the vehicle and ω_{max} is the maximum angular velocity of the antenna. In practical cases, the values of x_0, y_0 is not accurate and has error. The GPS error, is modelled as $e_{GPS}\ log\mathcal{N}(3,1)$ [2], to make the process error-tolerant. If the predicted position after time Δ is given by $(x, y) \leftarrow f(x_0 + e_{GPS_x}, y_0 + e_{GPS_y}, v_x, v_y)$, the prediction with correct information is $(x_c, y_c) \leftarrow f(x_0, y_0, v_x, v_y)$. We assess the error in the (x, y) w.r.t. (x_c, y_c) as the relative distance

$$e_p = \sqrt{(x_c - x)^2 + (y_c - y)^2} \tag{3}$$

A. Determining the minimum beam width

To define the simulation environment, we position the antenna at origin, and set the highway width to 4 m, the beam coverage= 100 m and maximum vehicle speed =27 m/s (roughly 100 km/h). The road length spans from x= -70 m to x= 70 m and the vehicles are assumed as point objects.

Simulation is performed keeping the GPS update time delay t_{delay} fixed. For a given t_{delay} the minimum required

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beamwidth θ_{min} is different. For increasing t_{delay} , θ_{min} also increases.$\theta_{min} = |\theta_p - \theta_r|$ where θ_p is the LOS angle of the predicted position of the vehicle and θ_r is the LOS angle of the real position of the vehicle at a given time.

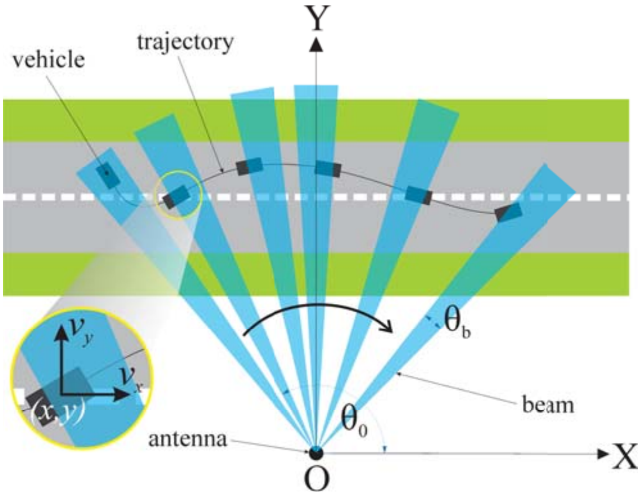

Fig. 1. Illustration of the considered scenario

B. Algorithm

Algorithm 1 Simulation Algorithm

while vehicle can be in LOS range of the antenna **do**
  if vehicle is in LOS range of antenna **then**
    Vehicle transmits positional information
    Motion prediction at antenna side
    Antenna performs beam steering
    GPS update time delay
  end if
end while

III. Results

In Fig. 2, we analyse $\bar{\theta}_{min} = \frac{1}{T}\int_0^T \theta_{min} \cdot dt$, over a range of t_{delay} . We calculate the directivity of the beam as $D = 2/(1 - \cos(\theta))$ and try to find the optimum range of time delay to set the beam width accordingly.

IV. Conclusion

The work is in its early stages and we aim to improve the simulation technology used and also consider other traffic flow models for a comparative study. In effect, what can be understood from the current results is that the higher is the latency in information update, the lower is the trajectory tracking accuracy and hence a wider beamwidth is required.

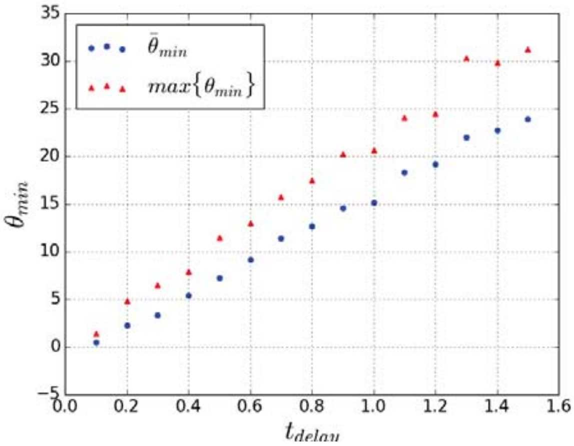

Fig. 2. Beam width variation with delay time

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